

# Emergent Digital Technologies

## Decision Trees

et al

1 de octubre de 2026

# Plan

Overview

A simple recipe

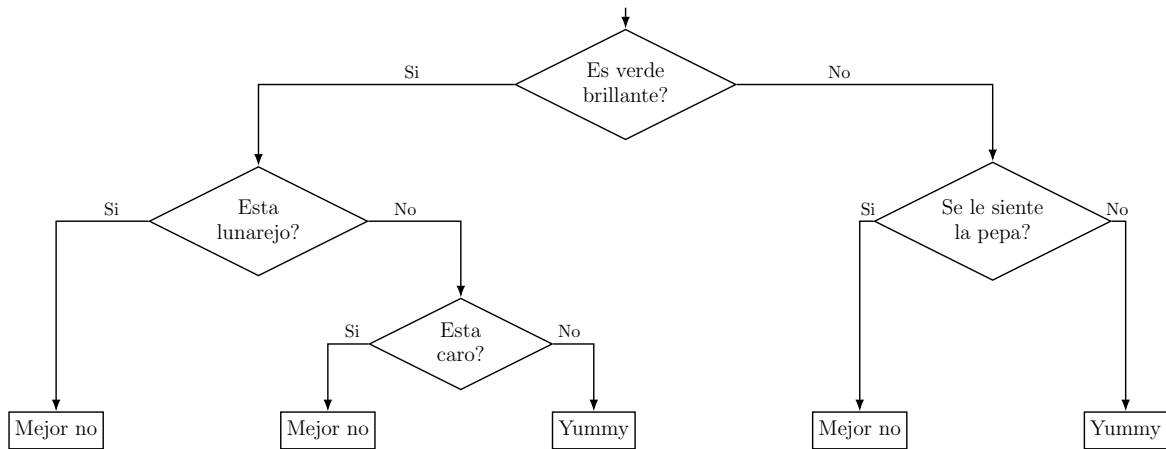
Foundation concepts

An example

# How to buy an avocado?



# How to buy an avocado...



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# 1. Measure the “Mess” (Entropy)

Before you start, look at your target (e.g., “Should buy?”) and see how mixed the data is.

- ▶ **Entropy** measures the level of disorder or uncertainty.
- ▶ **Formula:**  $H(S) = - \sum p_i \log_2(p_i)$
- ▶ **Maximum Mess:** If the data is  $\frac{50}{50}$  “Yes” and “No,” your Entropy is **1.0**.
- ▶ **Perfectly Clean:** If the data is all “Yes” (or all “No”), your Entropy is **0**.

## 2. Test Your “Clues” (Information Gain)

Pick a feature (like “Color” or “Price”) and pretend to split the data by it.

- ▶ Calculate the entropy of the new groups you created.
- ▶ Subtract this from your original mess.
- ▶ The Result = **Information Gain (IG)**.
- ▶ **Formula:**  $IG(S, A) = H(S) - \sum \frac{|S_v|}{|S|} H(S_v)$
- ▶ It tells you how much “cleaner” the data became because of that clue.

### 3. Pick the Winner (The Root)

Compare the Information Gain of every feature. The one with the highest score becomes the node.

- ▶ **The Root Node:** This is your first and most important split.
- ▶ **Leaf Nodes:** If a branch becomes 100 % “Yes” or “No,” stop there!
- ▶ **Internal Nodes:** If a branch is still messy, you need to keep digging and find the next best clue.

## 4. Rinse and Repeat (Recursion)

For any branch that isn't a Leaf yet, repeat Steps 1–3 using only the data that traveled down that specific path.

Repeat until:

- ▶ Every branch ends in a **Leaf** (0 Entropy).
- ▶ You **run out of features** to test.
- ▶ The tree reaches a **maximum depth** you've pre-set (to prevent it from getting too complex).

## Pro-Tip: When to Stop?

In an informal setting, we usually stop when every leaf is “pure”.

### Real-World Research

You often stop the tree early through **Pruning**. This prevents the model from “memorizing” the noise in the data (*overfitting*<sup>1</sup>) rather than learning the actual underlying patterns.

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<sup>1</sup>Have a look at “[What is overfitting?](#)” by IBM.

# Plan

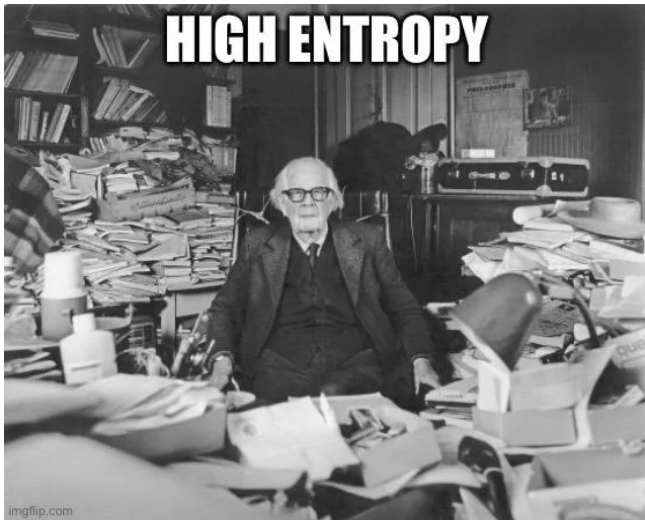
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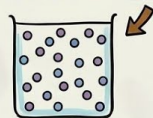
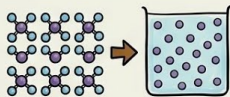
An example

# What is Entropy?



# WHAT IS ENTROPY?

## THE SECOND LAW OF THERMODYNAMICS



MOLECULAR  
DISORDER

WHITEBOARD LECTURE  
EASY EXPLANATION!



NEAT & ORDERED



**\*\*MAX DISORDER\*\***  
**\*\*INCREASED  
ENTROPY!\*\***



ARROW OF TIME  
- BREAKING EGG



MESSY  
DESK

What is Entropy? The Second Law of Thermodynamics Explained by Afzaal Chemist

# Understanding Entropy: The Measure of Disorder

**Entropy**  $H(S)$  quantifies the uncertainty or impurity within a dataset  $S$ . In the context of Decision Trees, it determines how “mixed” our target classes are.

## The Mathematical Formula

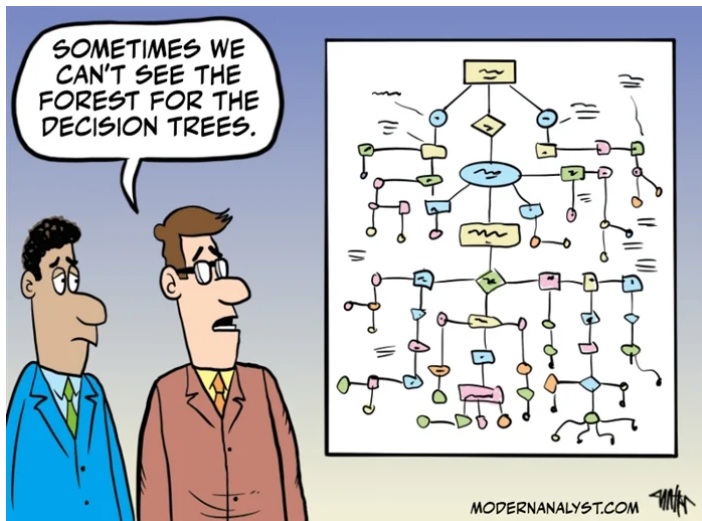
For a set with  $c$  classes, entropy is defined as:

$$H(S) = \sum_{i=1}^c -p_i \log_2(p_i)$$

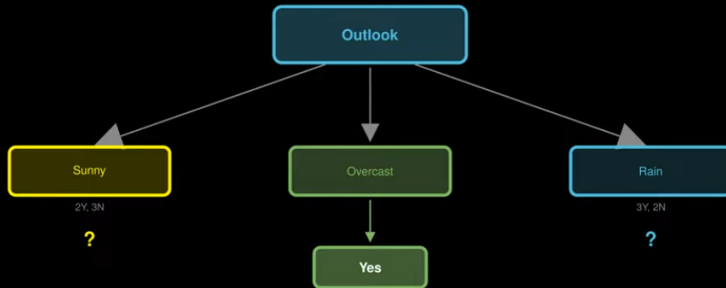
Where  $p_i$  is the proportion of samples belonging to class  $i$ .

- ▶ **High Entropy (1.0):** Data is perfectly split (e.g., 50 % Yes, 50 % No). Uncertainty is at its maximum.
- ▶ **Low Entropy (0.0):** Data is perfectly “pure” (e.g., 100 % Yes). There is zero uncertainty.
- ▶ **Goal:** We use splits to move from high entropy at the root to low entropy at the leaves.

# What is Information Gain?



## After the First Split



Apply information gain recursively to each impure branch

Information Gain Explained — Module 2 Ep 3 by Nipun Batra

# Information Gain: Choosing the Best Split

**Information Gain (IG)** measures the reduction in Entropy achieved by partitioning the data according to a specific attribute. It identifies which “clue” provides the most clarity.

## The Mathematical Formula

The Information Gain for a set  $S$  and attribute  $A$  is:

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v)$$

Where:

- ▶  $H(S)$  is the Entropy before the split.
- ▶  $\frac{|S_v|}{|S|}$  is the weight of the subset  $v$  resulting from the split.
- ▶  $H(S_v)$  is the Entropy of that subset.
- ▶ **The Selection Rule:** The attribute with the **highest** Information Gain is selected as the decision node.
- ▶ **Interpretation:** High IG means the attribute successfully organized the data into more homogeneous (pure) groups.

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# Introduction to the “Play Tennis” Dataset

The Play Tennis dataset is a classic example used to demonstrate the induction of Decision Trees via Entropy and Information Gain. It consists of 14 instances characterized by four environmental attributes.

## Target Variable:

- ▶ **PlayTennis:**  $\{Yes, No\}$

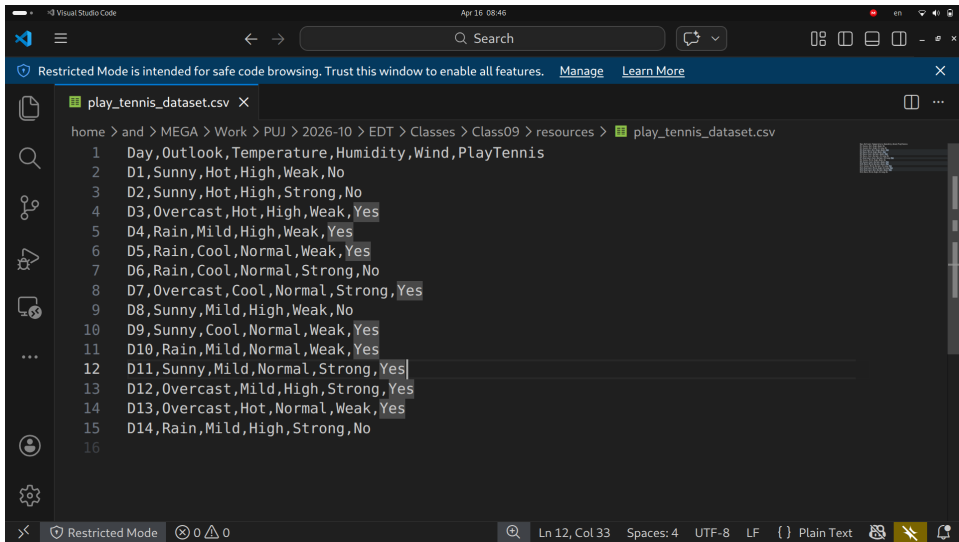
## Predictor Attributes:

- ▶ **Outlook:**  $\{Sunny, Overcast, Rain\}$
- ▶ **Temperature:**  $\{Hot, Mild, Cool\}$
- ▶ **Humidity:**  $\{High, Normal\}$
- ▶ **Wind:**  $\{Strong, Weak\}$

## Dataset Characteristics

- ▶ **Total Samples:** 14
- ▶ **Class Distribution:** 9 Yes / 5 No
- ▶ **Goal:** Predict if conditions are suitable for playing tennis based on weather observations.

# the “Play Tennis” Dataset



```
play_tennis_dataset.csv
home > and > MEGA > Work > PUJ > 2026-10 > EDT > Classes > Class09 > resources > play_tennis_dataset.csv
1 Day,Outlook,Temperature,Humidity,Wind,PlayTennis
2 D1,Sunny,Hot,High,Weak,No
3 D2,Sunny,Hot,High,Strong,No
4 D3,Overcast,Hot,High,Weak,Yes
5 D4,Rain,Mild,High,Weak,Yes
6 D5,Rain,Cool,Normal,Weak,Yes
7 D6,Rain,Cool,Normal,Strong,No
8 D7,Overcast,Cool,Normal,Strong,Yes
9 D8,Sunny,Mild,High,Weak,No
10 D9,Sunny,Cool,Normal,Weak,Yes
11 D10,Rain,Mild,Normal,Weak,Yes
12 D11,Sunny,Mild,Normal,Strong,Yes
13 D12,Overcast,Mild,High,Strong,Yes
14 D13,Overcast,Hot,Normal,Weak,Yes
15 D14,Rain,Mild,High,Strong,No
16
```

Restricted Mode

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## Step 1: Calculating Total Entropy $H(S)$

Before splitting, we calculate the entropy of the entire dataset  $S$ .

$$H(S) = - \left( \frac{9}{14} \right) \log_2 \left( \frac{9}{14} \right) - \left( \frac{5}{14} \right) \log_2 \left( \frac{5}{14} \right)$$

- ▶  $P(Yes) = 9/14 \approx 0,643$
- ▶  $P(No) = 5/14 \approx 0,357$
- ▶ **Result:**  $H(S) \approx 0,940$

## Step 2: Testing the First Attribute (Outlook)

We evaluate how much “Outlook” reduces the total entropy. Outlook has three values: Sunny, Overcast, and Rain.

- ▶ **Sunny:** (2 Yes, 3 No)  $\rightarrow H \approx 0,971$
- ▶ **Overcast:** (4 Yes, 0 No)  $\rightarrow H = 0$  (Perfectly Pure!)
- ▶ **Rain:** (3 Yes, 2 No)  $\rightarrow H \approx 0,971$

Next, we calculate the weighted average of these entropies to find the “remaining mess”.

### Step 3: Information Gain for Outlook

Subtract the weighted entropy of the split from the original  $H(S)$ .

$$\text{Weighted Avg} = \left( \frac{5}{14} \times 0,971 \right) + \left( \frac{4}{14} \times 0 \right) + \left( \frac{5}{14} \times 0,971 \right) \approx 0,693$$

$$IG(S, \text{Outlook}) = 0,940 - 0,693 = \mathbf{0,247}$$

## Step 3: Information Gain for Humidity

To evaluate **Humidity**, we split the 14 samples (9 Yes, 5 No) into **High** and **Normal** groups.  
Baseline Entropy  $H(S) \approx 0,940$ .

- ▶ **High Humidity:** 7 samples (3 Yes, 4 No)

$$H(\text{High}) = -\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7} \approx 0,985$$

- ▶ **Normal Humidity:** 7 samples (6 Yes, 1 No)

$$H(\text{Normal}) = -\frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} \approx 0,592$$

## Information Gain Calculation

$$IG(S, \text{Hum}) = 0,940 - \left[ \frac{7}{14}(0,985) + \frac{7}{14}(0,592) \right]$$

$$IG(S, \text{Hum}) = 0,940 - 0,7885 = \mathbf{0,151}$$

## Step 3: Information Gain for Wind

We evaluate the **Wind** attribute by splitting the 14 samples (9 Yes, 5 No) into **Weak** and **Strong** groups. Baseline Entropy  $H(S) \approx 0,940$ .

- ▶ **Weak Wind:** 8 samples (6 Yes, 2 No)

$$H(\text{Weak}) = -\frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8} \approx 0,811$$

- ▶ **Strong Wind:** 6 samples (3 Yes, 3 No)

$$H(\text{Strong}) = -\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6} = 1,0$$

## Information Gain Calculation

$$IG(S, \text{Wind}) = 0,940 - \left[ \frac{8}{14}(0,811) + \frac{6}{14}(1,0) \right]$$
$$IG(S, \text{Wind}) = 0,940 - 0,892 = \mathbf{0,048}$$

## Step 3: Information Gain for Temperature

We evaluate **Temperature** by splitting the 14 samples (9 Yes, 5 No) into **Hot**, **Mild**, and **Cool** groups. Baseline Entropy  $H(S) \approx 0,940$ .

- ▶ **Hot:** 4 samples (2 Yes, 2 No)  $\rightarrow H = 1,0$
- ▶ **Mild:** 6 samples (4 Yes, 2 No)  $\rightarrow H \approx 0,918$
- ▶ **Cool:** 4 samples (3 Yes, 1 No)  $\rightarrow H \approx 0,811$

### Information Gain Calculation

$$IG(S, \text{Temp}) = 0,940 - \left[ \frac{4}{14}(1,0) + \frac{6}{14}(0,918) + \frac{4}{14}(0,811) \right]$$

$$IG(S, \text{Temp}) = 0,940 - 0,911 = \mathbf{0,029}$$

## Step 4: Selecting the Root Node

We repeat the calculations for all other attributes:

- ▶  $IG(S, \text{Outlook}) = 0,247$
- ▶  $IG(S, \text{Humidity}) = 0,151$
- ▶  $IG(S, \text{Wind}) = 0,048$
- ▶  $IG(S, \text{Temperature}) = 0,029$

Decision

**Outlook** has the highest Information Gain. It becomes our **Root Node**.

## Step 5: Expanding the “Sunny” Branch

When Outlook is **Sunny**, the data is still mixed (2 Yes, 3 No). We must split again using only this subset.

### Subset Features (Outlook = Sunny):

- ▶ Humidity: (High  $\rightarrow$  3 No) and (Normal  $\rightarrow$  2 Yes)
- ▶ Wind: (Weak  $\rightarrow$  1 Yes, 2 No) and (Strong  $\rightarrow$  1 Yes, 1 No)

## Step 6: Final Split for the Sunny Branch

Evaluate Humidity on the “Sunny” subset ( $H_{subset} = 0,971$ ).

- ▶ **High Humidity:** All 3 are “No” ( $H = 0$ ).
- ▶ **Normal Humidity:** All 2 are “Yes” ( $H = 0$ ).

$$\text{Weighted Avg} = \left(\frac{3}{5} \times 0\right) + \left(\frac{2}{5} \times 0\right) = 0$$

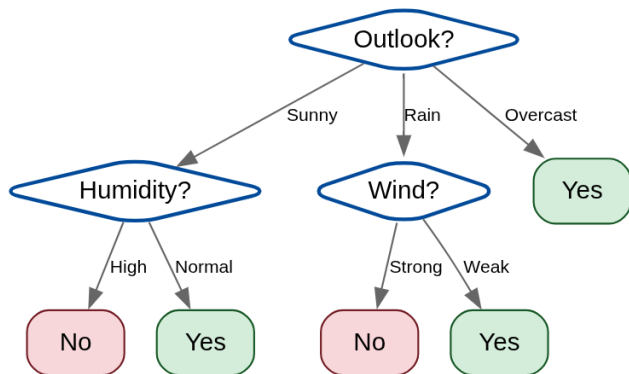
$$IG(\text{Sunny Subset, Humidity}) = 0,971 - 0 = \mathbf{0,971}$$

Humidity provides a perfect split for the Sunny branch.

## Step 8: Final Decision Tree Map

### Final Logic:

1. **Root:** Outlook
2. **If Overcast:** Leaf node  $\rightarrow$  **Yes**.
3. **If Sunny:** Split by **Humidity**.
  - ▶ High  $\rightarrow$  **No**.
  - ▶ Normal  $\rightarrow$  **Yes**.
4. **If Rain:** Split by **Wind**.
  - ▶ Weak  $\rightarrow$  **Yes**.
  - ▶ Strong  $\rightarrow$  **No**.



# Takeaways: Decision Tree Induction

1. **The Construction Recipe:** A systematic four-step process that uses Entropy and Information Gain to build tree structures through recursive splitting.

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3. **Information Gain for Feature Selection:** The primary “clue finder” that identifies the best feature to split on by calculating the reduction in entropy achieved.

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1. **The Construction Recipe:** A systematic four-step process that uses Entropy and Information Gain to build tree structures through recursive splitting.
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4. **Iterative Partitioning and Termination:** Tree growth continues recursively until branches reach zero-entropy “leaf nodes” or the model hits a pre-defined depth limit.
5. **Generalization through Pruning:** A critical technique that stops tree growth early to prevent “overfitting,” ensuring the model learns patterns rather than memorizing noise.